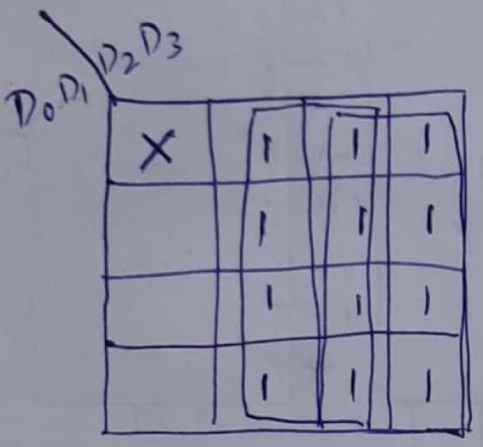
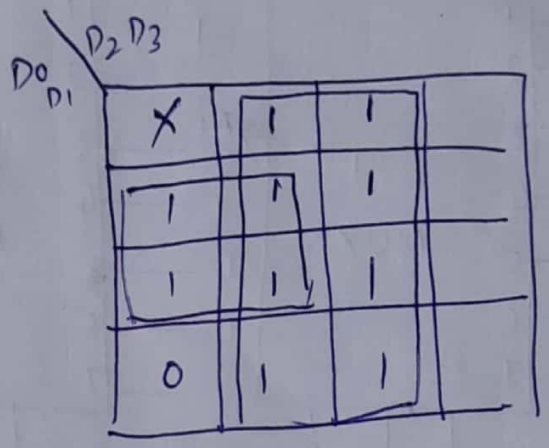
Priority Encoder :- Design a priority encoder for the truth table shown below. Assume that D_3 has highest priority & D_0 has least priority.

i/p				o/p		
D_0	D_1	D_2	D_3	X	Y	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

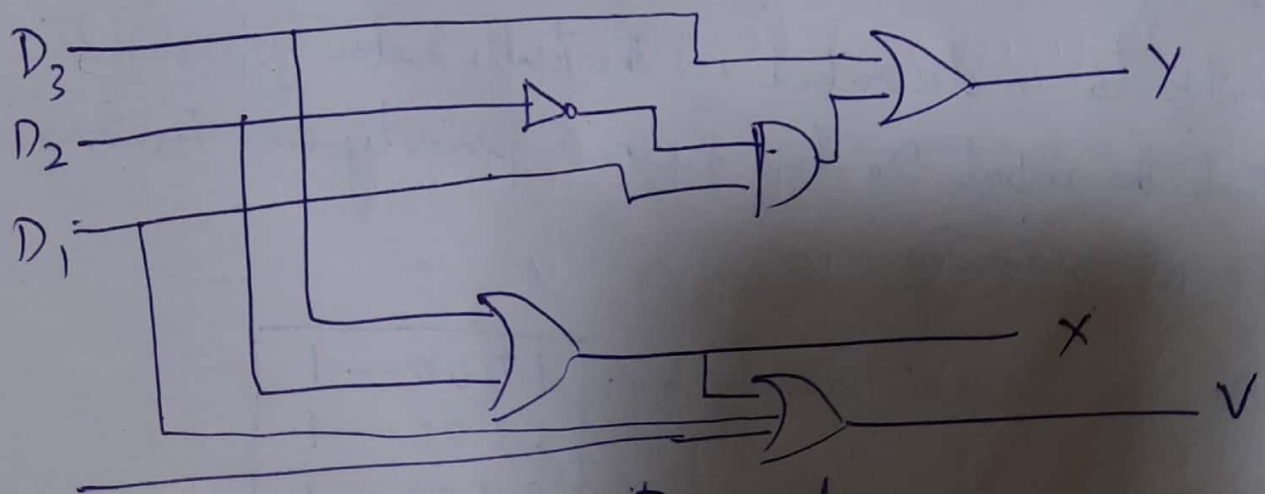


$X = D_2 + D_3$



$Y = D_3 + D_1 D_2'$

$V = D_0 + D_1 + D_2 + D_3$

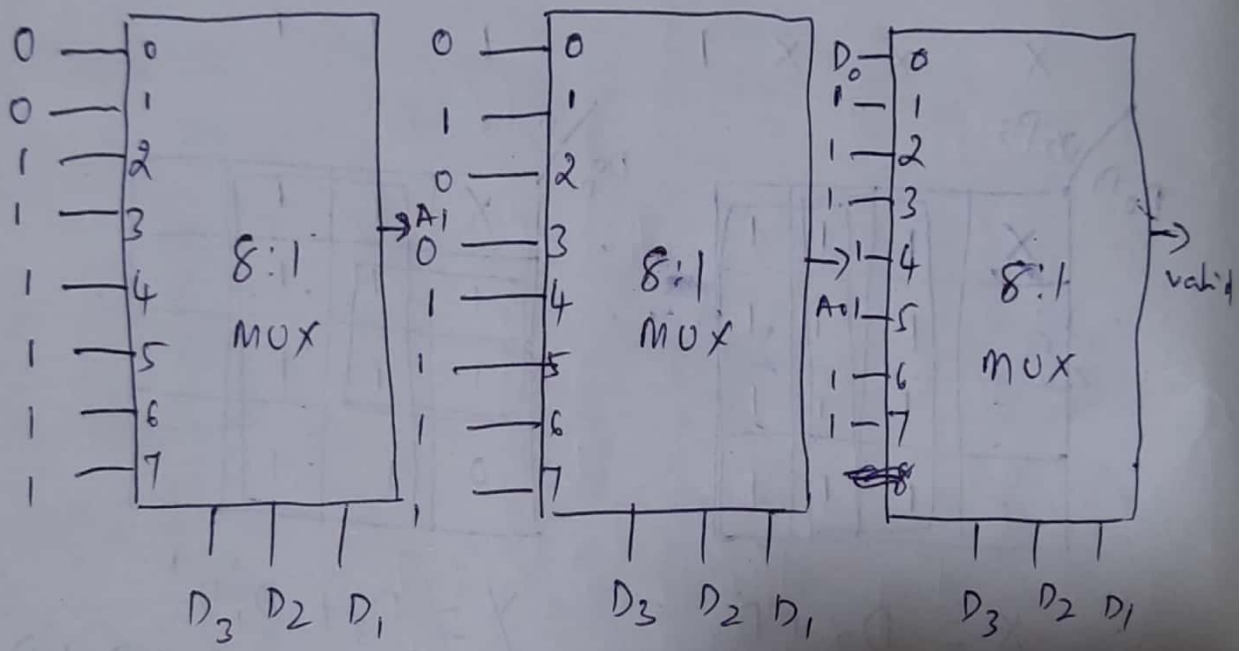


4 - input priority encoder

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28 plement a 4:2 priority encoder using only 8:1 MUX & inverters.

D_3	D_2	D_1	D_0	A_1	A_0	valid
0	0	0	0	x	x	0
0	0	0	1	0	0	1
0	0	1	x	0	1	1
0	1	x	x	1	0	1
1	x	x	x	1	1	1



1) Design a 4-input priority encoder with inputs and outputs as described in the table below. please note that the input D_0 has highest priority and D_3 has lowest priority.

D_0 has highest priority — 1000 — 1
1100 — 1
0100 — 1
0000 — x

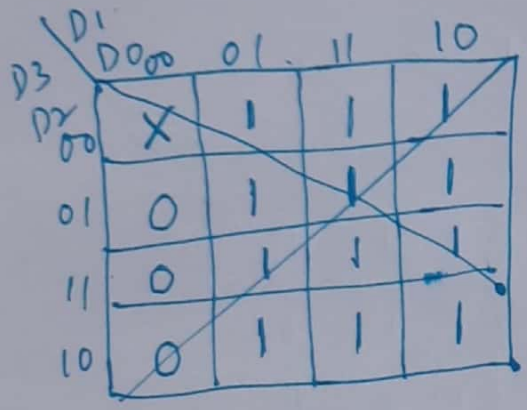
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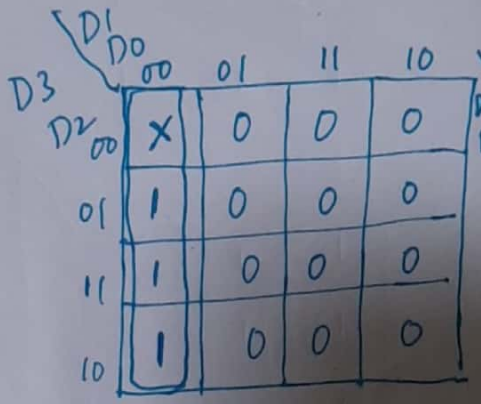
I/P

D_3	D_2	D_1	D_0
0	0	0	0
1	0	0	0
X	1	0	0
X	X	1	0
X	X	X	1

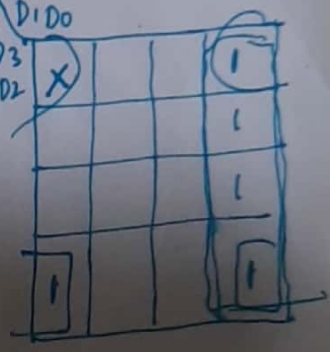
O/P

X	Y	V
X	X	0
0	0	1
0	1	1
1	0	1
1	1	1



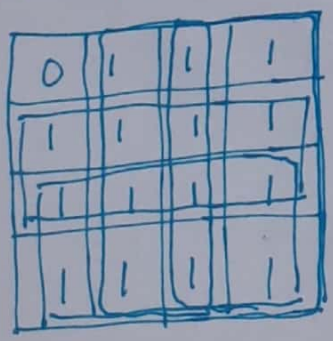


$X = D_1' D_0'$



$Y = D_2' D_0' + D_1 D_0'$

$V = D_0 + D_1 + D_2 + D_3$



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